

Glossary: Network Security Techniques

Welcome! This alphabetized glossary contains many of the terms you'll find within this course. This comprehensive glossary also includes additional industry-recognized terms not used in course videos. These terms are important for you to recognize when working in the industry, participating in user groups, and participating in other certificate programs.

Term	Definition
Access Control List (ACL)	A set of rules used to determine whether a source IP address or service is allowed into the network.
Application Files	Executable files and associated resources necessary for running applications to ensure correct and secure software operations.
Application-Level Firewalls	Firewalls that operate at the application layer of the OSI model and inspect packet payloads to provide granular control over network traffic based on specific application data.
Artificial Intelligence (AI)	Use of machines and algorithms to simulate human intelligence, improving threat detection by analyzing patterns and anomalies in security data.
Automated Remediation	The process of automatically addressing non-compliance issues and restoring network access for legitimate users.
Behavioral Analysis	Monitors endpoint and network behaviors to detect deviations from regular activity, helping to identify zero-day and advanced persistent threats.
Class of Service (CoS)	A method used to prioritize network traffic by marking it, enhancing the management of network quality and performance.
Cloud Security	Measures and technologies used to protect data and applications stored in cloud environments, integrated into XDR for comprehensive security.
Data Loss Prevention (DLP)	A set of tools and processes used to monitor, detect, and block unauthorized data transfers, protecting sensitive information from internal and external threats.
Destination IP Address	The IP address of the target device or server that a network packet is intended to reach.
Destination Network Address Translation (NAT)	Modifies the destination IP address of incoming packets, often used to direct traffic to a specific internal server or service.
Endpoint Detection and Response (EDR)	A set of cybersecurity tools and practices designed to detect, investigate, and respond to suspicious activities on endpoints like desktops, laptops, servers, and mobile devices, providing continuous monitoring and analysis of endpoint activities.
File Integrity Monitoring (FIM)	A cybersecurity process that continuously verifies the integrity of system and application software files by comparing them to a trusted baseline to detect unauthorized changes.
Firewall	A security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules.
Host-Based IDS (HIDS)	Intrusion detection systems installed on individual hosts or devices that monitor system logs, file modifications, and process activities to detect potential intrusions or unauthorized access attempts.
Host-Based Intrusion Prevention System (HIPS)	Deployed on individual hosts to protect the device at the endpoint level by blocking suspicious activities.
Intrusion Detection System (IDS)	A network security technology designed to monitor network traffic for suspicious activities and potential threats.
Intrusion Prevention System (IPS)	Builds upon the capabilities of IDS by not only detecting threats but also taking proactive steps to block or mitigate them, such as dropping malicious packets and blocking offending IP addresses.
Network Address Translation (NAT)	Provides extra security layer between private and public networks by modifying network address information in IP headers and masking internal IP addresses from external networks.
Network-Based IDS (NIDS)	Intrusion detection systems to monitor network-wide traffic patterns to flag anomalies and potential threats, such as unauthorized access attempts or unusual traffic patterns.
Port Address Translation (PAT)	Also known as NAT overload, maps multiple private IP addresses to a single public IP address with different port numbers for each session, commonly used in home networks.
Static NAT	Network Address Translation that maps a single private IP address to a single public IP address, providing a fixed external address for internal devices.
Synergistic Security Approach	The combined use of firewall filters, IDS, and IPS systems creates a comprehensive defense mechanism against cyber threats, where firewalls enforce security policies, and IDS and IPS systems provide additional layers of dynamic protection.
VLAN	Virtual Local Area Network is used to segment network traffic and manage data flow within a network.



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